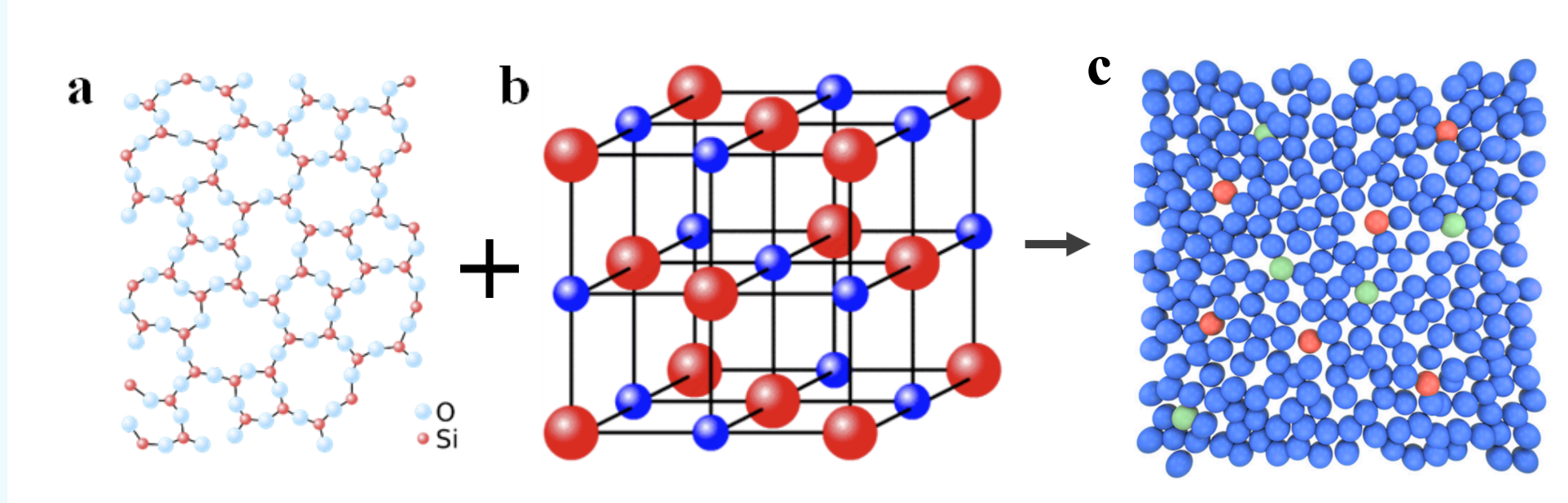


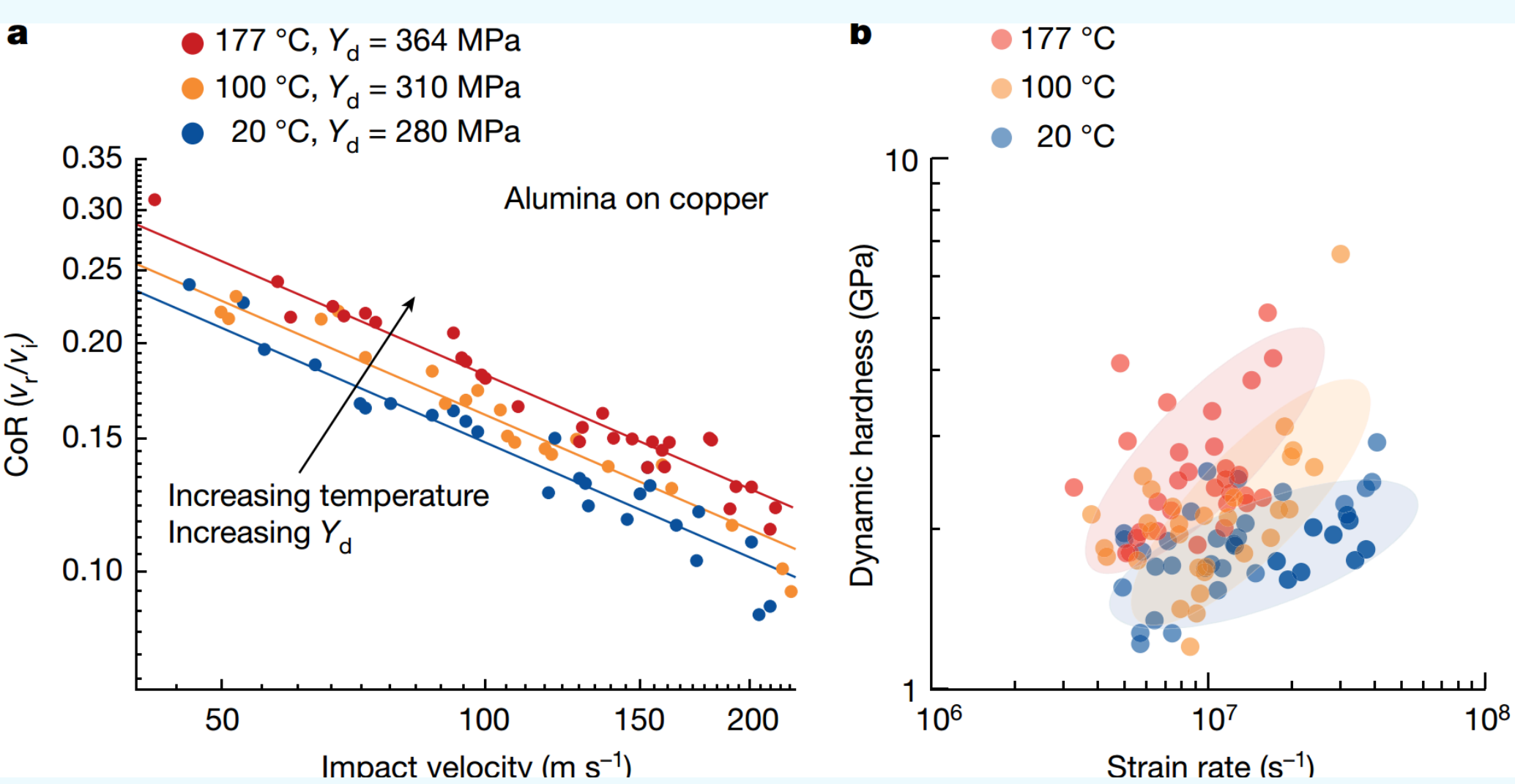
Introduction

- Unlike crystalline metals with long-range order, Bulk Metallic Glasses (BMGs) possess a disordered, "liquid-like" atomic structure.
- This unique configuration grants BMGs exceptional strength and potential for extreme structural applications, such as high strength at room temperature



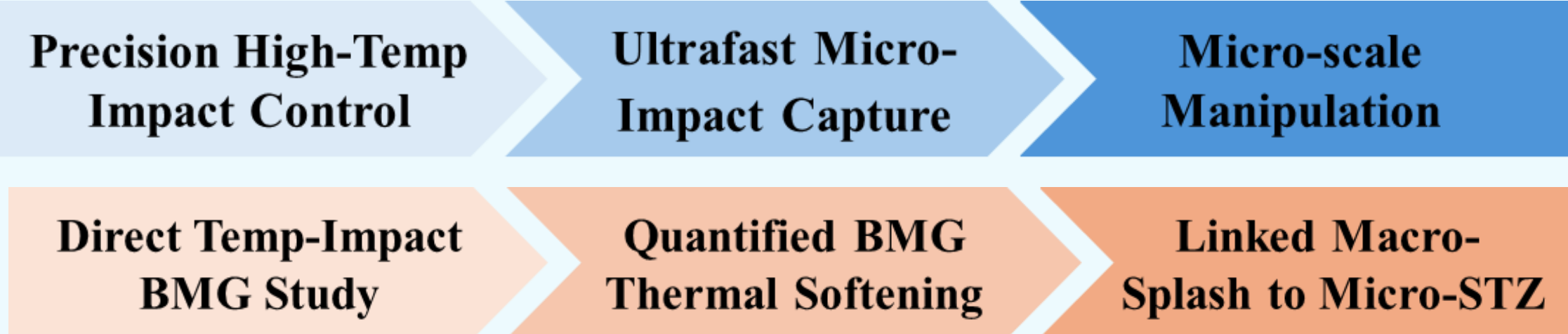
Previous Studies

- under moderate strain rates, increasing temperature typically softens metals
- under high strain rates, the dynamic yield strength of pure metals both increases with rising temperature due to the effect of phonon-drag strengthening

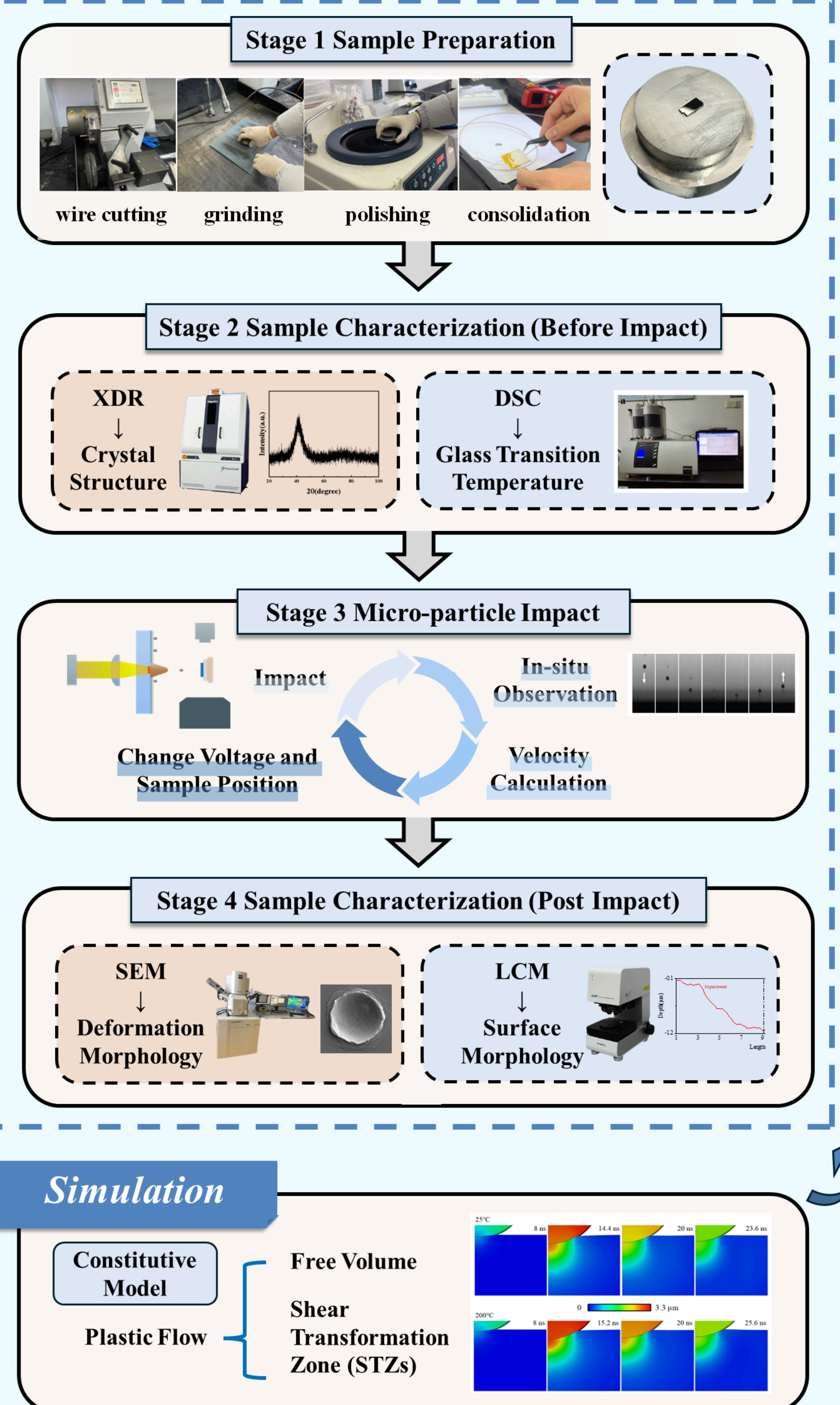


Research Question and Design

- How does temperature influence the energy dissipation of BMGs under high-velocity micro-particle impact?
- What is the underlying mechanism of this macroscopic physical phenomena?



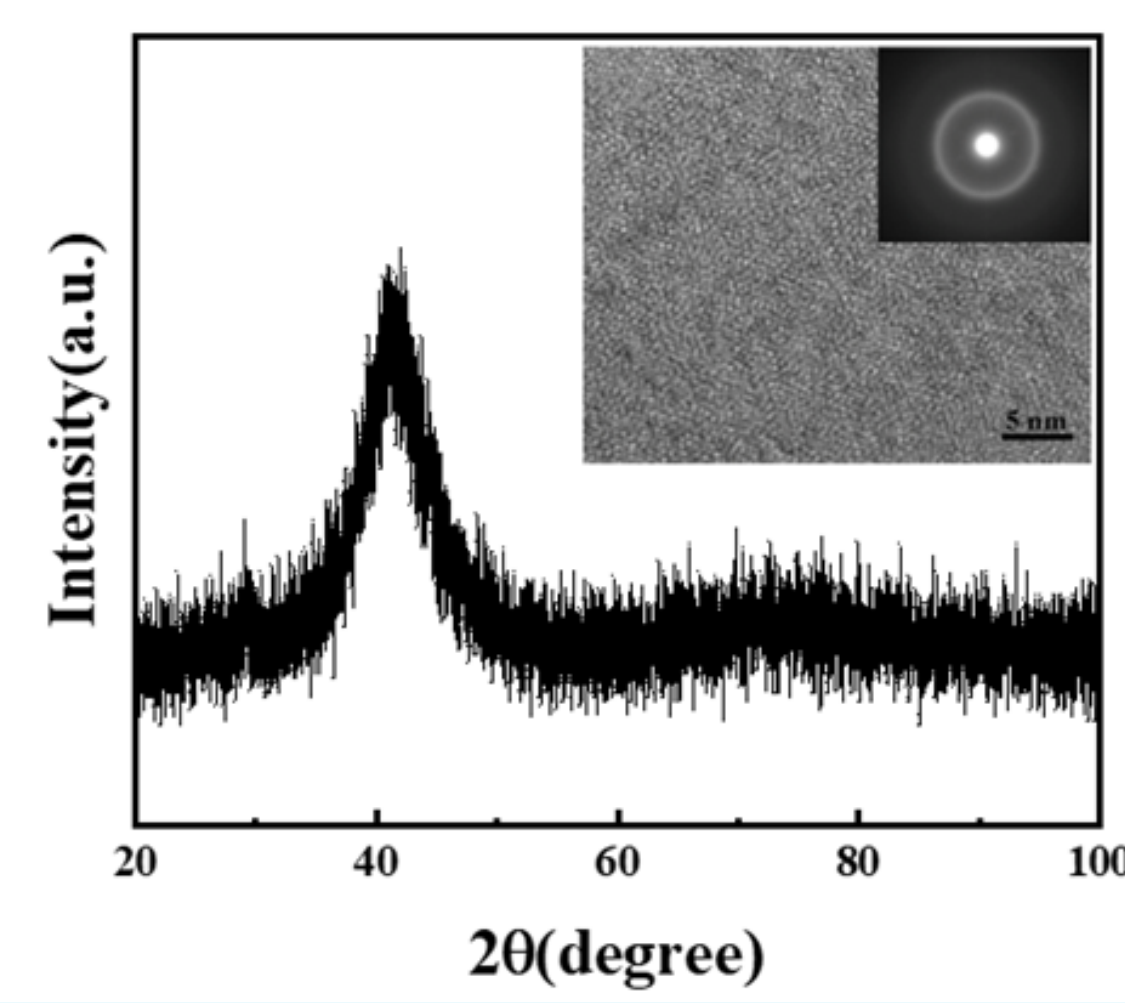
Experiment



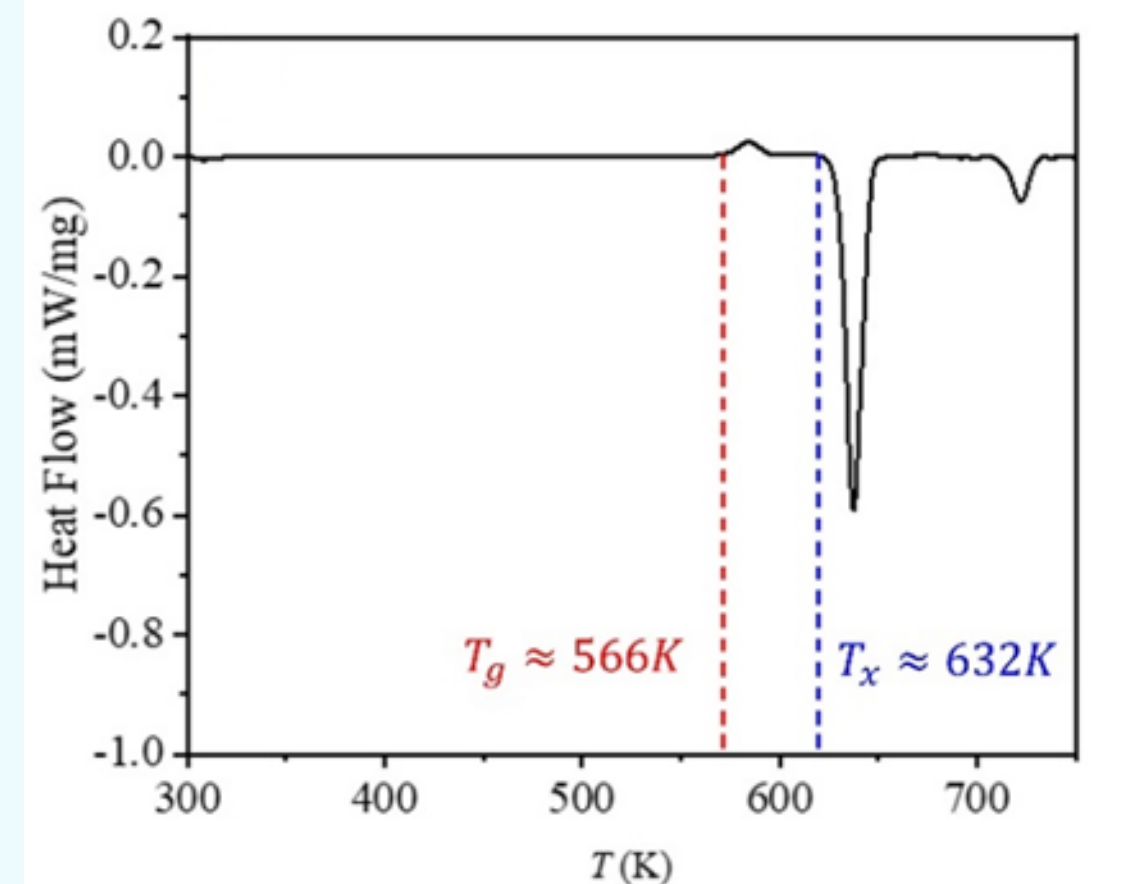
Reference

- [1] Scholze, H. (2012). Glass: nature, structure, and properties, P45-48. Springer Science & Business Media.
- [2] Wang, W. H. (2011). A Brief History of Metal Glass Research. Physics, 40(11), 701-709.
- [3] Guo, Y. C., Wang, Z. X. (1984). Amorphous State Physics, Beijing: Science Press
- [4] Wang, W. H., Dong, C., Shek, C. H. (2004). Bulk metallic glasses. Materials Science and Engineering: R: Reports, 44(2-3), 45-89.
- [5] Dowding, L., Schuh, C. A. (2024). Metals strengthen with increasing temperature at extreme strain rates. Nature, 630(8015), 91-95.
- [6] Sakino, K. (2000). Transition in rate controlling mechanism of FCC metals at very high strain rates and high temperatures. Le Journal de Physique IV, 10(PR9), Pr9-57.
- [7] Kuno, M. (2001). Thermodynamics of the Pd₄₃Ni₄₆Cu₁₁P₂₀ metallic glass-forming alloy. Oregon State University.
- [8] Wu, C. Y., Li, L. Y., Thornton, C. (2003). Rebound behaviour of spheres for plastic impacts. International journal of impact engineering, 28(9), 929-946.

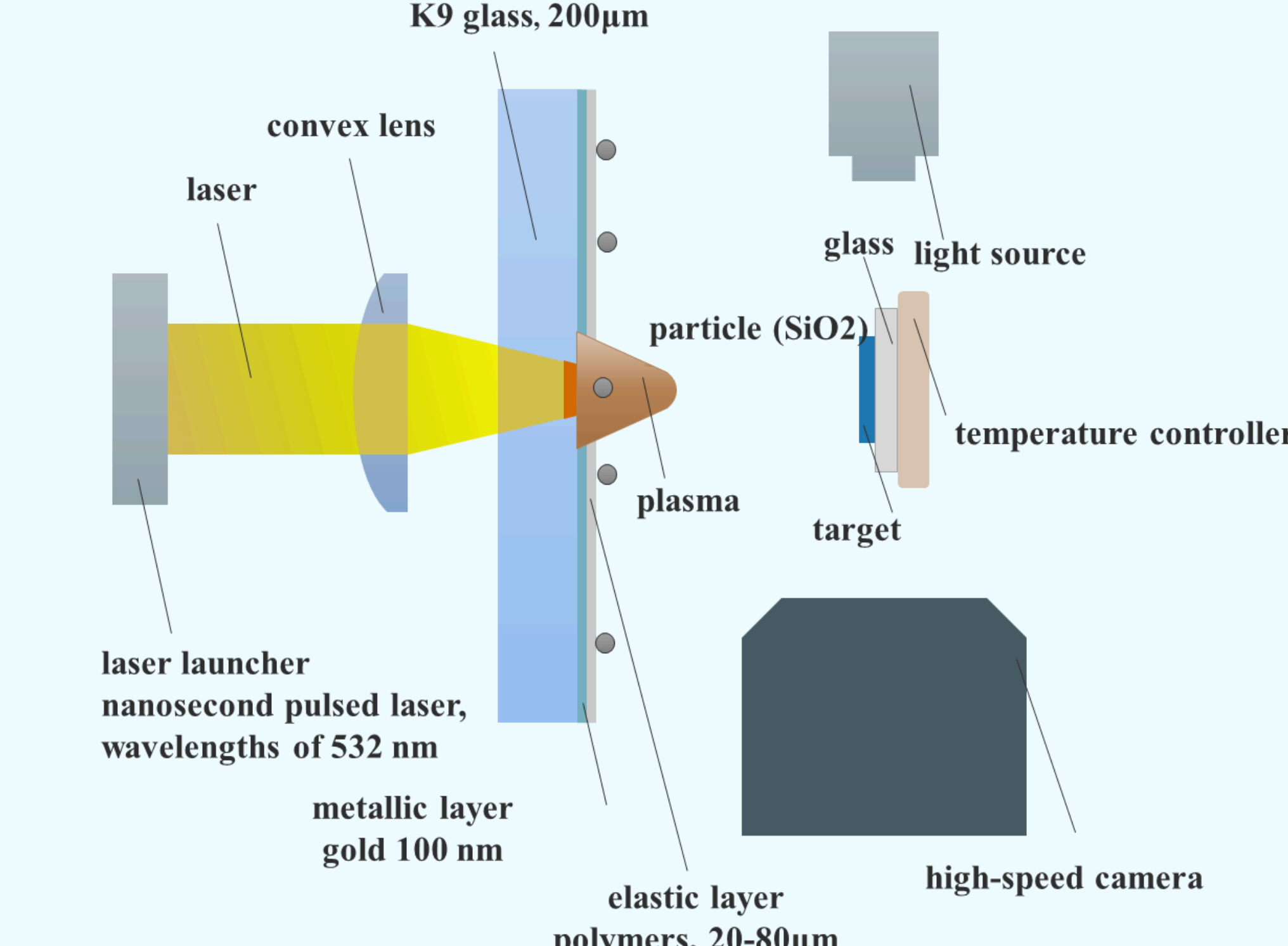
Methodology



XRD pattern with the TEM observation



DSC results indicating T_g and T_x

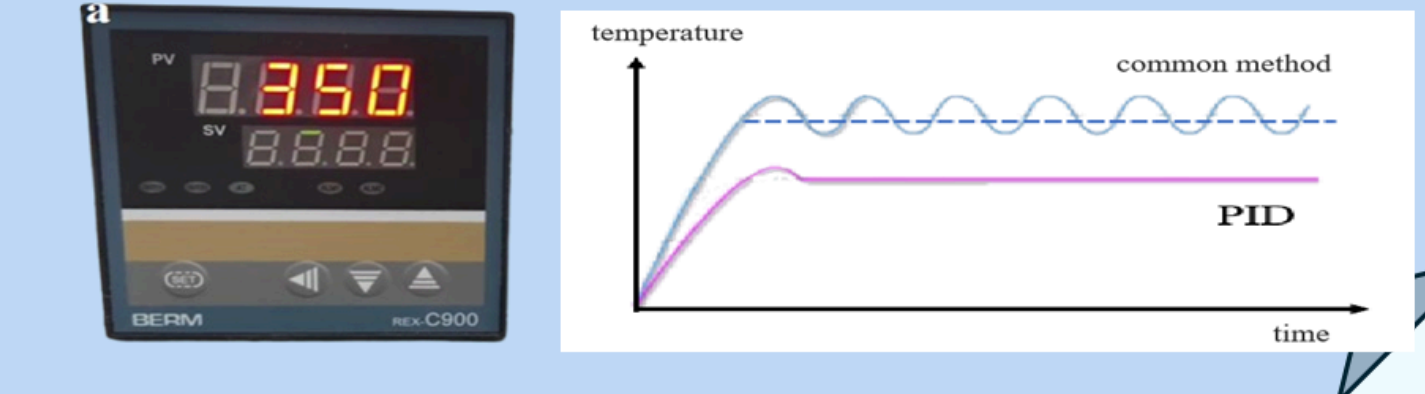


Experiment Equipment 1: LIPIT

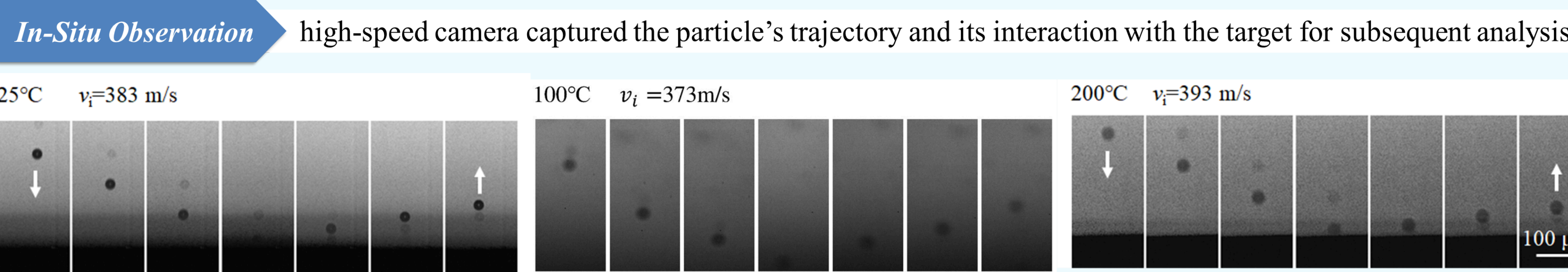
- Laser-Induced Particle Impact Tests
- Target: BMG Pd₄₃Ni₄₆Cu₁₁P₂₀, chosen for high stability
- Projectile: 30 μm SiO₂ micro-particles
- velocity: 70 m/s ~ 700 m/s

Experiment Equipment 2: PID temperature controller

- Precision to ±1K
- Utilizes Proportional-Integral-Derivative (PID) technology
- 25 °C, 100 °C and 200 °C

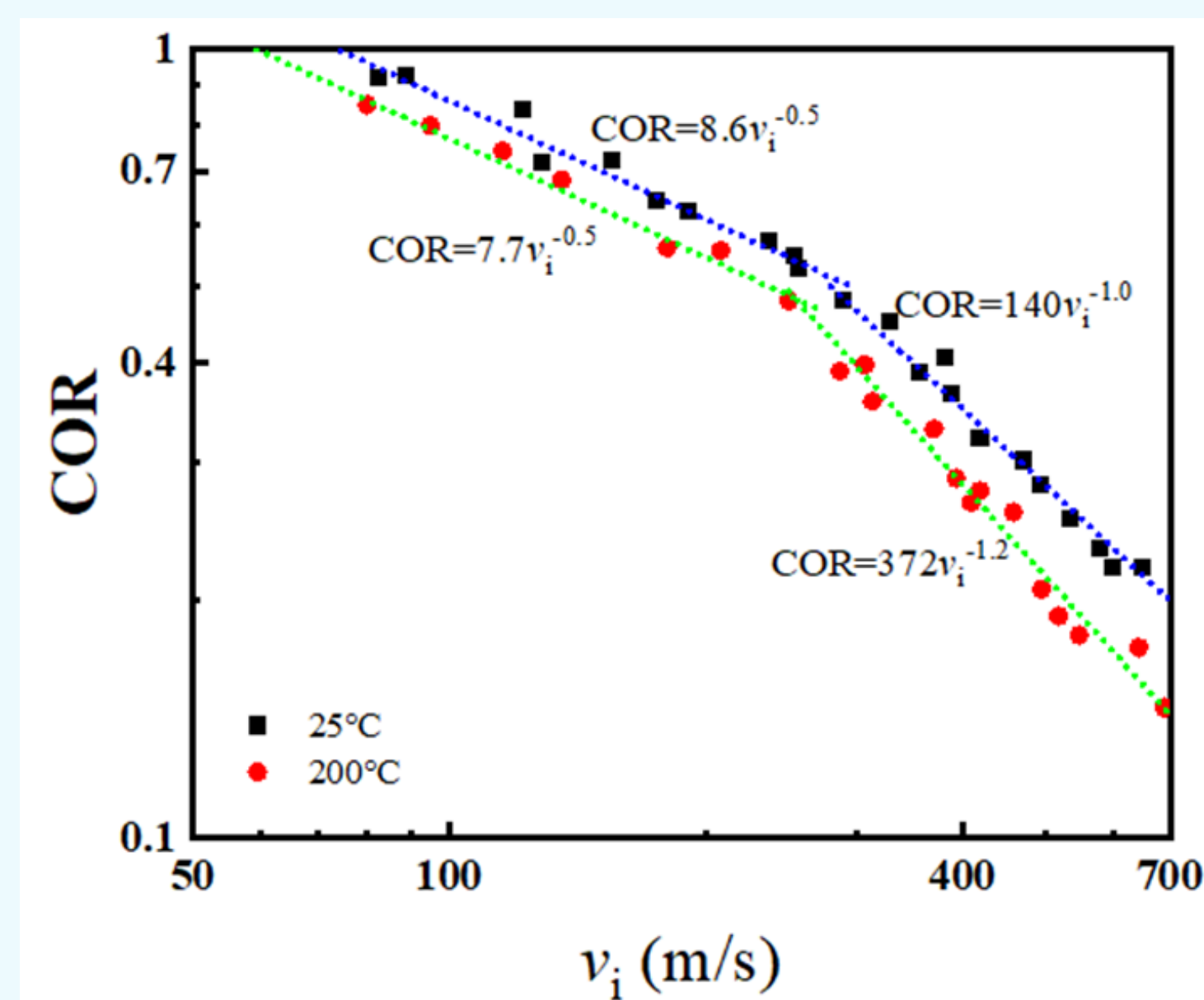


Results and Discussion



- under all temperatures, the post-impact particle displacements are substantially smaller than the pre-impact displacements
- rebound velocity of the particles is notably lower than their impact velocity, showing the system undergoes inelastic collisions
- no particle adhesion was observed in either temperature group, the particle remains a ball shape after impact

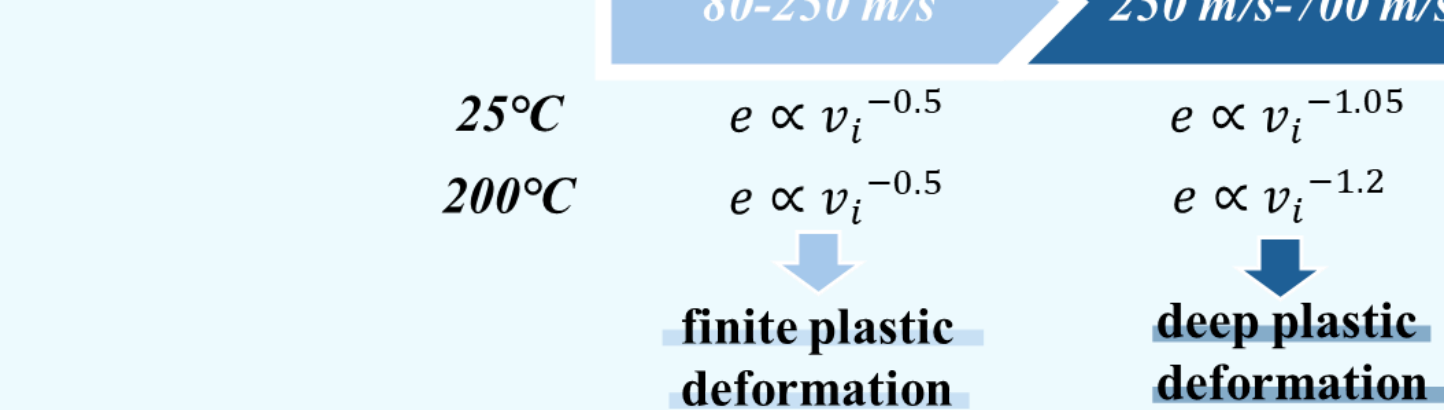
Coefficient of Restitution ratio of rebound velocity to impact velocity, reflects energy dissipation during collision



Coefficient of restitution for micro-particles impact on glassy targets under different temperatures

$$COR = v_r / v_i$$

- As impact velocity increases ↑, COR tends to decrease ↓, indicating greater kinetic energy loss through plastic deformation and stress wave propagation

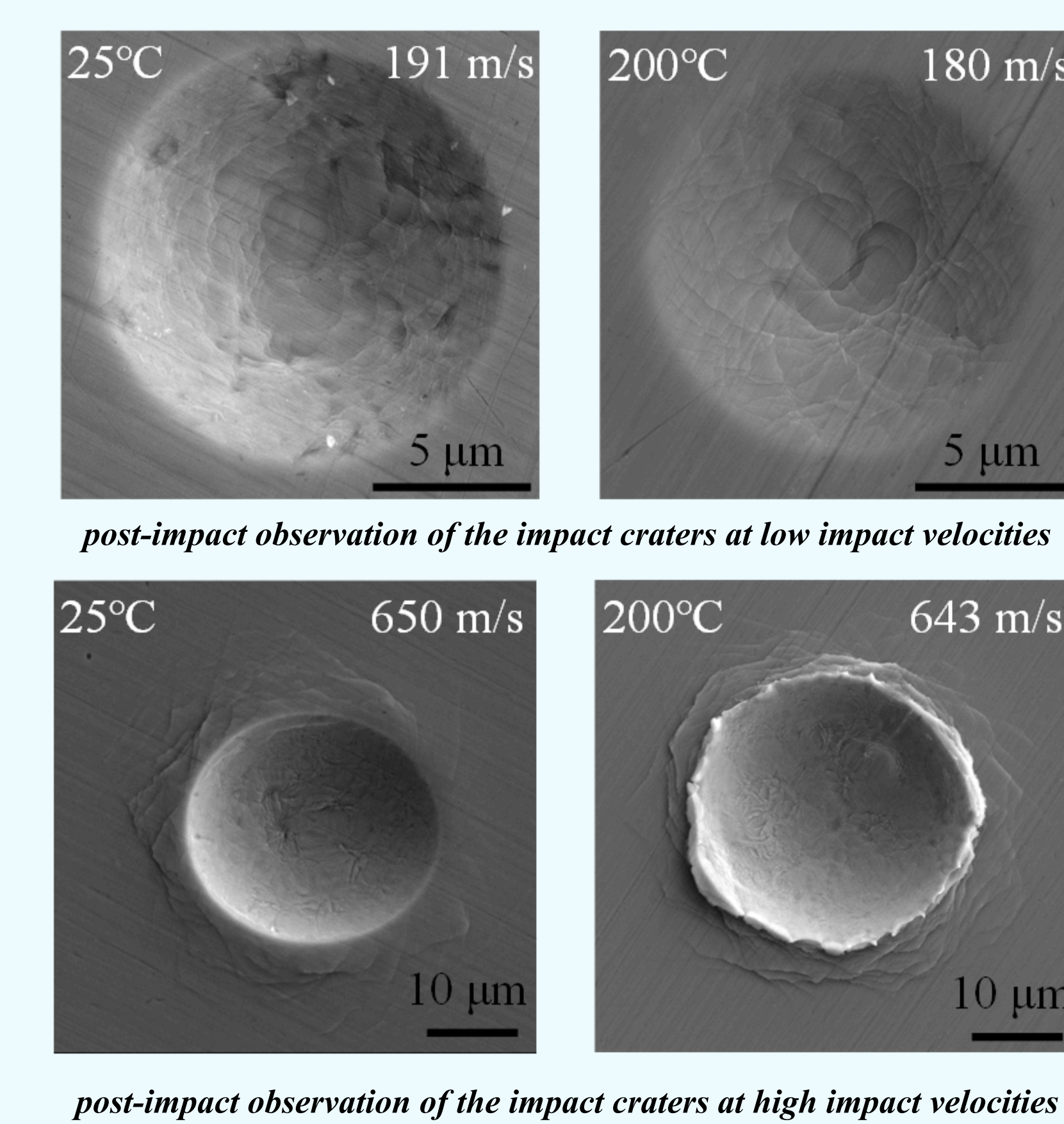


- COR at 200 °C are consistently lower than that at 25 °C, indicating material dissipates more energy

better energy dissipation capability at high temperature.

better anti-penetration resistance at low temperature

Surface Morphology impact crater features are further observed by SEM



post-impact observation of the impact craters at low impact velocities

post-impact observation of the impact craters at high impact velocities

- Both Targets: circular craters, relatively smooth 'flat area' lies in the middle of the crater; annular bands on the inner wall of craters

due to the initiation and development of shear bands in amorphous targets (for energy dissipation)

- Both have regular crater shape, with smooth edges and no pile-up → maintain a solid phase
- little difference between different temperatures

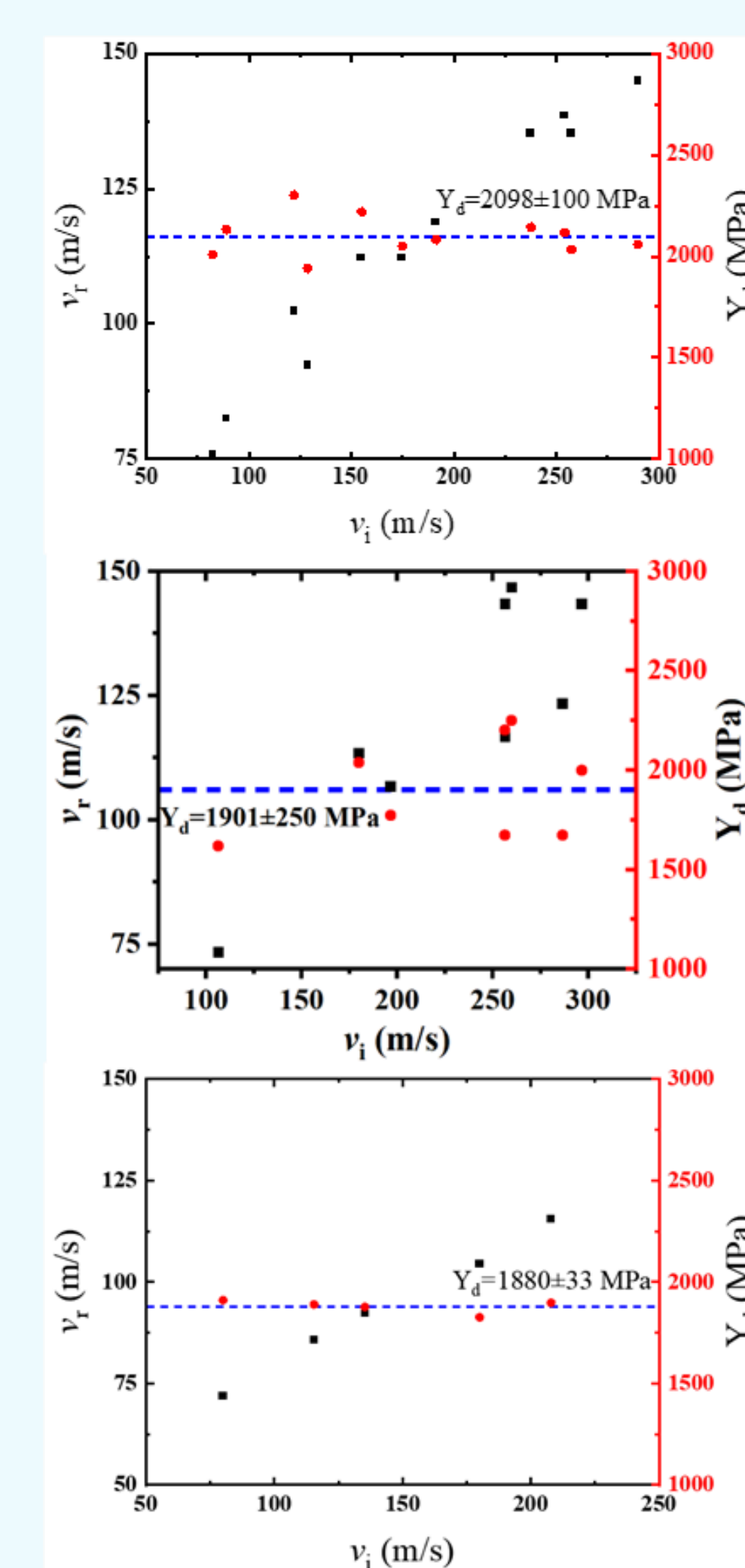
- Higher temperature the BMG exhibit a larger deformation → more remarkable energy dissipation and a lower COR value

- Both samples exhibit pile-up at the edge of the crater, while higher and has a larger volume at higher temperature; pile-up is regular at lower temperature, while highly irregular, has an outward-splash pattern at higher temperature

characteristics of highly viscous fluid caused by the adiabatic temperature rise

Dynamic Yield Strength

critical stress threshold at which a material transitions from elastic to permanent plastic deformation



Dynamic yield strength of targets under different temperatures

When samples are in finite plastic regime, according to FEA results by Chuan-yu Wu et al. and Johnson and Li et al, the formula for dynamic yield strength is:

$$\frac{1}{E^*} = \frac{1 - \mu_p^2}{E_p} + \frac{1 - \mu_t^2}{E_t}$$

$$Y_d = \left(\frac{\rho E^* v_r^4}{26 v_i^2 \alpha^4} \right)^{\frac{1}{3}}$$

- 2089±110 MPa at 25 °C, 1901 MPa at 100 °C and 1825±65 MPa at 200 °C

- Y decreases when temperature increases

- decreases by 10.39% from 25 °C to 200 °C

- BMG exhibits a better energy dissipation mechanism under high temperature while has a better impact resistance under low temperature (conforms the previous analysis)

Simulation

FEM simulation was carried out by using the coupled shear transformation zones (STZs)-free volume constitutive model.

$$\begin{aligned} \dot{\epsilon}_{ij}^{eff} &= \exp\left(-\frac{1}{2}\right) \exp\left[\frac{1}{2}\left(\frac{\sigma_{ij}}{\sigma_0}\right)\right] \left[\frac{1}{2} \left(\frac{\sigma_{ij}}{\sigma_0} \right) - \frac{1}{2} \cosh\left(\frac{\sigma_{ij}}{\sigma_0}\right) \right] \frac{\sigma_{ij}}{\sigma_0} \\ \dot{\epsilon}_{ij} &= \epsilon_{ij}(\dot{\epsilon}_{ij}^{eff}) \\ \dot{\epsilon} &= 2\dot{\epsilon}_{ij}^{eff} - \phi(\dot{\epsilon}_{ij}^{eff}) \dot{\epsilon} \\ \dot{\epsilon} &= \phi(\dot{\epsilon}_{ij}^{eff}) \left(\frac{1}{2} - \dot{\epsilon} \right) \\ \dot{\epsilon} &= \phi(\dot{\epsilon}_{ij}^{eff}) - \exp\left(-\frac{1}{2}\right) \exp\left[\frac{1}{2}\left(\frac{\sigma_{ij}}{\sigma_0}\right)\right] \left(\frac{1}{2} - \dot{\epsilon} \right) \end{aligned}$$

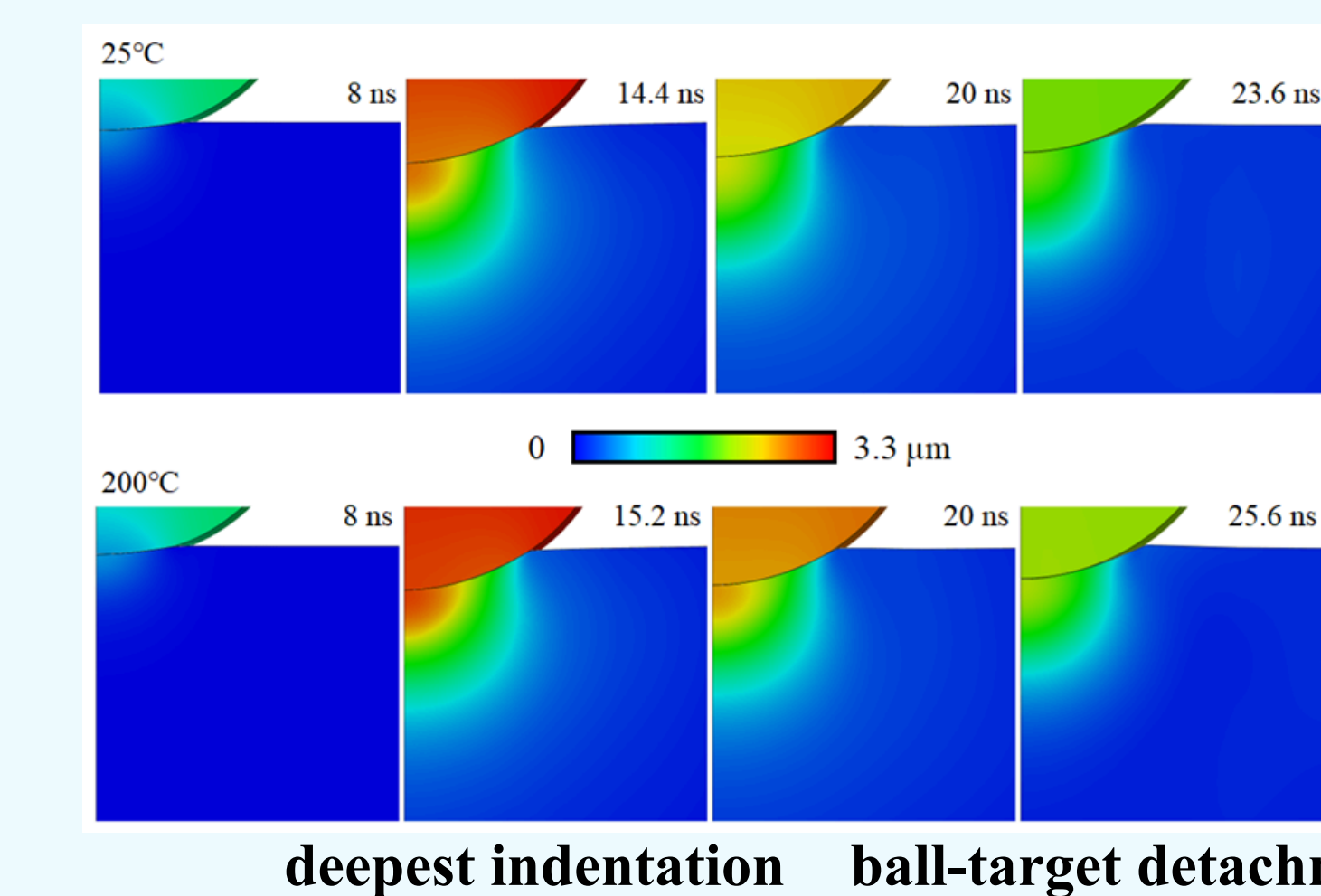


The FEM model for micro-particle impacting on the BMG

Comparison between experiment and simulation: the displacement of the micro-particle impacting on the target

indentation progress

displacement cloud images of characteristic time points



- longer interaction time with the target at higher temperatures

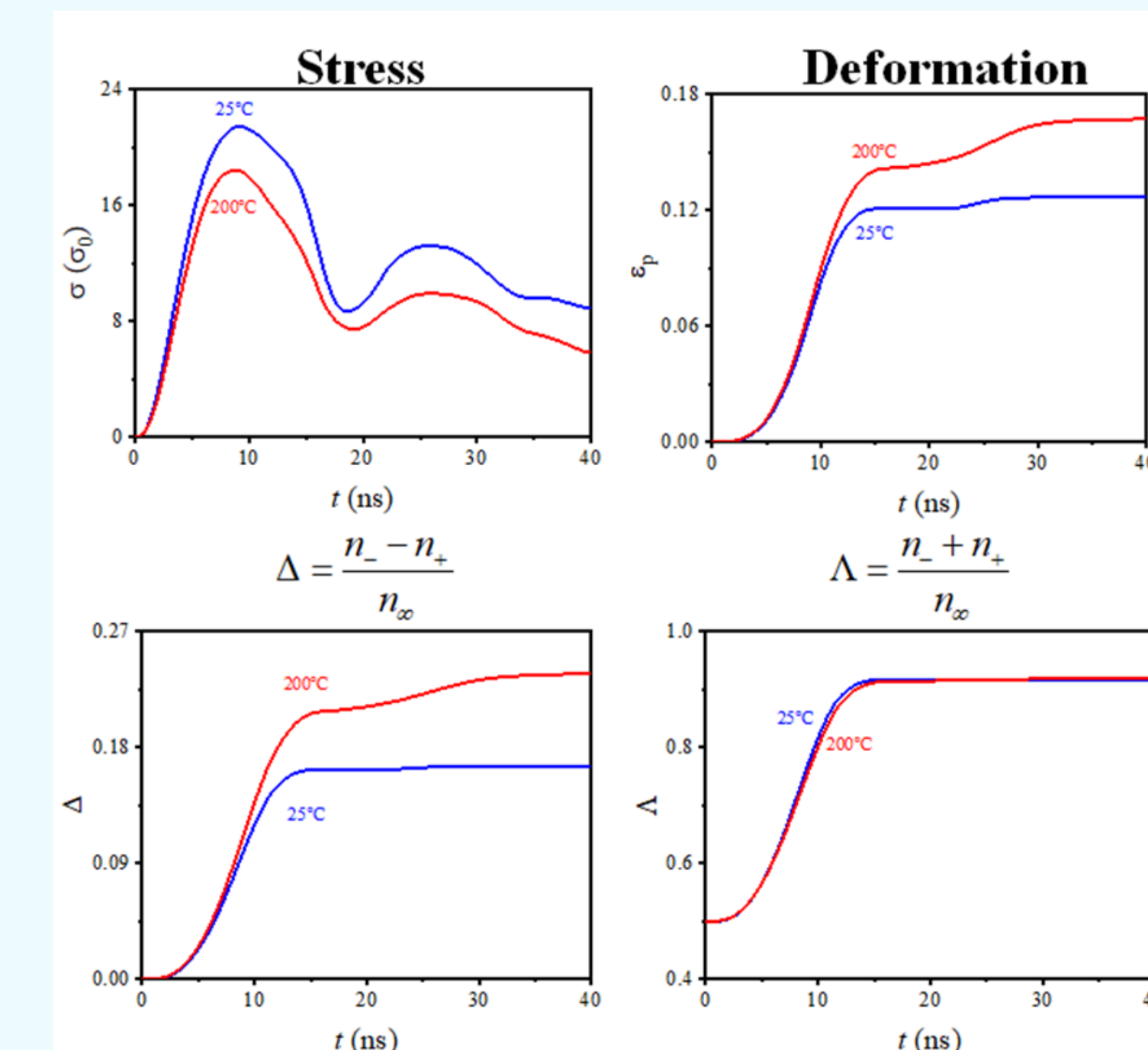
longer energy transfer

- greater deformation (cloud image color is deeper at high T)

deepest indentation ball-target detachment

Field Average Results

Simulated results of stress, equivalent plastic strain, difference/sum of the STZ number density



- stress: lower at high T
- eq plastic strain: rapid growth-equilibrium at low T; rapid growth-slow growth at high T
- difference in number density of STZs and eq plastic strain show same trend

high temperature promotes the shear transformations, resulting in greater plastic strain

Conclusion

- COR decreases as velocity and temperature increase, indicating enhanced energy dissipation at higher temperatures.
- At 200 °C, craters shift from solid-like to fluid-like "splash" patterns due to adiabatic heating near T_g
- Dynamic yield strength drops by 10.39% (from 25 °C to 200 °C), confirming significant thermal softening.
- High temperature prolongs impact interaction time and promotes the activation of Shear Transformation Zones (STZs).

Prospects

- Conduct experiments in a vacuum or inert gas environment to prevent oxidation, allowing testing above 200 °C to explore the glass transition temperature range
- Utilize ultrafast thermometry to precisely measure adiabatic heating during impact
- Evaluate different BMG compositions to engineer materials for specific extreme-environment applications, such as spacecraft shielding.

[9] Johnson, K. L. (1987). Contact mechanics. Cambridge university press.

[10] Spaepen, F. (1977). A microscopic mechanism for steady state inhomogeneous flow in metallic glasses. Acta metallurgica, 25(4), 407-415.

[11] Jiang, M. Q., Dai, L. (2007). Intrinsic correlation between fragility and bulk modulus in metallic glasses. Physical Review B—Condensed Matter and Materials Physics, 76(5), 054204.

[12] Jiang, M. Q., Jiang, S. Y., Dai, L. H. (2009). Inherent shear-dilatation coexistence in metallic glass. Chinese Physics Letters, 26(1), 016103.

[13] Argon, A. S. (1979). Plastic deformation in metallic glasses. Acta metallurgica, 27(1), 47-58.

[14] Sopu, D., Stukowski, A., Stoica, M., Seidino, S. (2017). Atomic-level processes of shear band nucleation in metallic glasses. Physical review letters, 119(19), 195503.

[15] Xu, B., Falk, M. L., Li, J. F., & Kong, L. T. (2018). Predicting shear transformation events in metallic glasses. Physical review letters, 120(12), 125503.

[16] Langer, J. S. (2001). Microstructural shear localization in plastic deformation of amorphous solids. Physical Review E, 64(1), 011504.

[17] Lemaître, A. (2002). Rearrangements and dilatancy for sheared dense materials. Physical Review Letters, 89(19), 195503.